



### Prevent spread indoors

Remap your ventilation and air filtration system to help prevent the coronavirus spread indoors. <http://digit.in/nov20-21>



### CBD lowers COVID-19 lung damage

CBD reduces COVID-19 related lung damage by boosting the levels of apelin, a natural peptide. <http://digit.in/nov20-22>

then. In the news channel coverage, the polarisation was elevated, but the politicisation remained at low levels.

Another study conducted by researchers from the University of Chicago has demonstrated that political partisanship has undermined the response to the virus. It is the politicisation of the pandemic, that pushes people to reject masks, not follow social distancing guidelines, or ignore lockdowns. Even though the experts recommended the wearing of masks as far back as April, the issue remains contentious because it has been politicised into one of individual freedom, with mostly those associated with the Republican Party refusing to wear masks. According to the authors of the study, the uncertainty and lack of consensus in the early stages of the pandemic, allowed political actors to craft politicised narratives, that had the end result of weakening the public compliance to the recommended guidelines. If the political parties had given out similar messages in the early stages of the pandemic, then it is likely that most people would have taken up to mask usage without it becoming a matter of such acute contention. The study found that political allegiance strongly defined mask use, local policy guidelines and advocacy was unable to offset the political effect, partisanship was the most important predictor of mask use irrespective of how prevalent the outbreak was or the local policies in place, and that when President Donald Trump unexpectedly used a mask, the sentiments towards mask use turned positive almost instantly. The study found that those areas that voted for Trump in 2016, also had the people most reluctant to use masks. The research indicates that polarised societies may be less effective at responding to future pandemics.

### UNDERSTANDING THE VIRUS

Caltech researchers are investigating what exactly the virus does within human cells, to cause the known

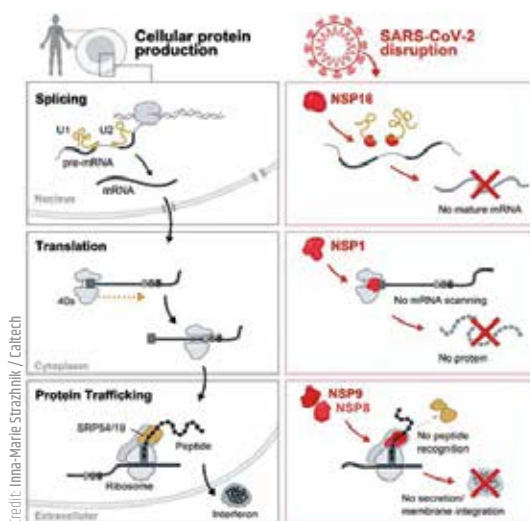
symptoms of breathlessness, cough, fever and fatigue. Viruses do not have the cellular machinery for self replication, and this is why they invade healthy cells so that they can use their organelles for replicating themselves. Mammalian cells have evolved elaborate defence mechanisms to prevent viruses from hijacking them, and one of these is a production of a group of proteins called interferon. The production of interferon happens after alien viral material is detected within the cell. The interferon is secreted by cells signals neighbouring cells of the infection by a virus, to halt the

being spliced and assembled, others prevent ribosomes from forming new proteins, and still others block protein transport within the cell. The research allows scientists to not only tackle the novel coronavirus better, but also establishes an approach to take on any RNA virus that may become a problem in the future.

Research from the University of Manchester has shown that white blood cells produced by the bone marrow, known as monocytes, have abnormal characteristics, which can be spotted in the early stages of the infection to identify which patients are likely to have a severe response to the virus. The study also indicates that the monocytes themselves can be the target of treatments for the disease. At this point of time, it is unknown if the abnormalities are formed in the bone marrow itself, or once the blood cells enter the bloodstream.

A study conducted on 5,000 patients by the University of Texas, Austin has found that the virus is accumulating mutations, which may be allowing it to spread faster. One of the mutations, called D614G, was also identified as a possible factor that helps

the virus spread faster, according to another study conducted in the UK. There are two factors contributing to the rapid evolution of the virus, natural genetic drift, as well as the immune response from the affected patients. In an initial survey, only 71 percent of the patients had the virus with the D614G, but 99.9 percent of the patients in the second wave of the infection had viruses with the mutation. The researchers have identified 285 mutations, though a majority of them have no impact on how quickly the virus can spread. Every time a cell is infected, the dice is rolled again, giving the opportunity for the virus to develop a new mutation. **4**



Normal production of proteins by cells on the left, and the disruptions to splicing, translation and protein trafficking on the right.

spread of the virus. Those with severe COVID-19 symptoms also have a subdued interferon response, which indicates that this response is necessary to contain the infection within the body, and that SARS-CoV-2 has a mechanism by which it can suppress the production of interferon. Researchers identified and tracked around 30 proteins produced by the virus, and what these proteins do within the cell. The proteins produced by the virus interfere at multiple stages with the mechanisms by which cells convert the information stored in DNA into proteins useful for the body, including interferon. Some proteins prevent mRNA from